Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

12 Oct, 2026 | Tutorial-7 | 3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
5. Submissions will not be accepted after the tutorial hours.

Hypothesis Testing

Q1. **One-Sample Mean Testing (Known Variance)** A high-speed rail facility tests composite suspension struts designed for a target mean frequency $\mu_0 = 150.0$ Hz ($\sigma = 3.5$ Hz). A sample of $n = 45$ struts yields $\overline{x} = 151.2$ Hz.

(a) Formulate hypotheses to test whether the true mean resonant frequency has shifted upward.

(b) Conduct the hypothesis test at $\alpha = 0.05$. State the test statistic, decision rule, and conclusion.

(c) If the true mean is $\mu_1 = 151.8$ Hz, compute the Type II error probability (β). ■

Table Values: $z_{0.05} = 1.645$, $z_{0.025} = 1.960$.

Q2. **Two-Sample Mean Comparison (Known Variances)** Two geothermal turbine blade alloy compositions undergo high-temperature creep tests ($\sigma_1 = 1.2$ mm/yr for Alloy-1, $\sigma_2 = 1.5$ mm/yr for Alloy-2). Independent sample trials yield:

Alloy-1: $n_1 = 50$, $\overline{x}_1 = 12.4$ mm/yr vs. Alloy-2: $n_2 = 60$, $\overline{x}_2 = 13.1$ mm/yr

(a) Set up hypotheses to test if Alloy-2 exhibits a significantly greater mean creep rate than Alloy-1.

(b) Test the hypothesis at $\alpha = 0.01$ and state your conclusion.

(c) Compute the p-value associated with the observed test statistic. ■

Table Values: $z_{0.01} = 2.326$, $z_{0.005} = 2.576$.

Q3. **One-Sample Mean Testing (Unknown Variance)** An environmental engineer monitors landfill leachate discharge where the regulatory limit is $\mu_0 = 25.0$ mg/L. A sample of $n = 15$ weekly water samples yields $\overline{x} = 26.8$ mg/L and $s = 3.2$ mg/L (normally distributed population).

(a) Formulate hypotheses to test if the true mean concentration exceeds the regulatory limit.

(b) Perform the hypothesis test at $\alpha = 0.05$, stating the T-statistic and degrees of freedom df.

(c) Construct the two-sided 95% confidence interval for μ and verify its alignment with a two-sided test. ■

Table Values: $t_{0.05,14} = 1.761$, $t_{0.025,14} = 2.145$.

Q4. **Two-Sample Independent t-Test (Equal Variances)** Two robotic welding protocols are evaluated for joint shear strength on structural steel frames (independent normal populations, equal variances):

$$\text{Protocol X: } n_1 = 12,\ \overline{x}_1 = 215.4 \text{ MPa},\ s_1 = 6.2 \text{ MPa}$$

$$\text{Protocol Y: } n_2 = 14,\ \overline{x}_2 = 219.0 \text{ MPa},\ s_2 = 5.8 \text{ MPa}$$

(a) State hypotheses to determine if Protocol Y provides a higher mean shear strength than Protocol X.

(b) Calculate the pooled sample variance s_p^2 and the resulting test statistic.

(c) Perform the test at $\alpha = 0.05$ and state the conclusion. ■

Table Values: $t_{0.05,24} = 1.711$, $t_{0.025,24} = 2.064$.

Q5. **Hypothesis Testing on Population Variance** An aerospace specification requires that turbine blade root variance must not exceed $\sigma_0^2 = 0.04$ mm^2. A sample of $n = 25$ blade roots yields $s^2 = 0.06$ mm^2 (normal distribution).

(a) Set up hypotheses to test whether the variance exceeds the strict quality threshold.

(b) Compute the test statistic and perform the hypothesis test at $\alpha = 0.05$.

(c) Determine the range of the p-value using the provided table boundaries. ■

Table Values: $\chi^2_{0.95,24} = 13.848$, $\chi^2_{0.05,24} = 36.415$, $\chi^2_{0.975,24} = 12.401$, $\chi^2_{0.025,24} = 39.364$.

Q6. **Chi-Squared Goodness-of-Fit Test** A laboratory records compressive strength ($n = 20$ concrete core samples). Raw measurements are given below:

Sample ID (1–10)	1	2	3	4	5	6	7	8	9	10
Strength (x_i)	36.1	41.5	39.8	42.1	40.5	37.4	43.0	39.1	41.0	38.8
Sample ID (11–20)	11	12	13	14	15	16	17	18	19	20
Strength (x_i)	42.5	40.2	39.5	41.8	38.0	40.9	42.2	39.0	41.2	40.0

Sample statistics: $\overline{x} = 40.23$ MPa, $s = 1.56$ MPa. Data is grouped into $k = 4$ intervals: $(-\infty, 38.5]$, $(38.5, 40.0]$, $(40.0, 41.5]$, and $(41.5, \infty)$.

(a) Determine the observed frequencies O_i for each interval from the raw data table.

(b) Using the z-table, compute interval standardized limits, theoretical probabilities, and expected frequencies E_i.

(c) State the formula for degrees of freedom df with m estimated parameters, and compute df $(k = 4, m = 2)$.

(d) Compute χ^2_{calc} and test at $\alpha = 0.05$ to determine if the normal model is acceptable. ■

Table Values: $\chi^2_{0.95,1} = 0.004$, $\chi^2_{0.05,1} = 3.841$, $\chi^2_{0.975,1} = 0.001$, $\chi^2_{0.025,1} = 5.024$; $\chi^2_{0.95,2} = 0.103$, $\chi^2_{0.05,2} = 5.991$, $\chi^2_{0.975,2} = 0.051$, $\chi^2_{0.025,2} = 7.378$; $\chi^2_{0.95,3} = 0.352$, $\chi^2_{0.05,3} = 7.815$, $\chi^2_{0.975,3} = 0.216$, $\chi^2_{0.025,3} = 9.348$.